Malignant pleural mesothelioma: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up

SUPPLEMENTARY MATERIAL

Supplementary Table S1. UICC version 8 staging definitions for T, N, and M descriptors of MPM

T (primary tumour)	
TX	Primary tumour cannot be assessed
T0	No evidence of primary tumour
T1	Tumour involves the ipsilateral parietal or visceral pleura only, with or without involvement of visceral, mediastinal, or diaphragmatic pleura
T2	Tumour involves the ipsilateral pleura (parietal or visceral pleura), with at least one of the following: • Invasion of diaphragmatic muscle • Invasion of lung parenchyma
T3	Tumour involves ipsilateral pleura (parietal or visceral pleura), with at least one of the following: • Invasion of endothoracic fascia • Invasion into mediastinal fat • Solitary focus of tumour invading soft tissues of the chest wall • Non-transmural involvement of the pericardium
T4	Tumour involves ipsilateral pleura (parietal or visceral pleura), with at least one of the following:

	• Chest wall, with or without associated rib destruction (diffuse or multifocal) • Peritoneum (via direct transdiaphragmatic extension) • Contralateral pleura • Mediastinal organs (oesophagus, trachea, heart, great vessels) • Vertebra, neuroforamen, spinal cord • Internal surface of the pericardium (transmural invasion with or without pericardial effusion)
N (regional lymph nodes)	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastases to ipsilateral intrathoracic lymph nodes (includes ipsilateral bronchopulmonary, hilar, subcarinal, paratracheal, aortopulmonary, paraesophageal, peridiaphragmatic, pericardial fat pad, intercostal, and internal mammary nodes)
N2	Metastases to contralateral intrathoracic lymph nodes. Metastases to ipsilateral or contralateral supraclavicular lymph nodes
M (distant metastasis)	
M0	No distant metastasis
M1	Distant metastasis

MPM, malignant pleural mesothelioma.[1]

Supplementary Table S2: UICC version 8 stage grouping for MPM

Stage	T	N	M
IA	T1	N0	M0
IB	T2, T3	N0	M0
II	T1, T2	N1	M0
IIIA	T3	N1	M0
IIIB	T1, T2, T3 T4	N2 Any N	M0 M0
IV	Any T	Any N	M1

MPM, malignant pleural mesothelioma. [1]

Supplementary Table S3: ESMO-MCBS table for new therapies/indications in malignant pleural mesothelioma

Therapy	**Disease setting**	**Trial**	**Control**	**Absolute survival gain**	**HR (95%CI)**	**QoL/ toxicity**	**ESMO-MCBS score[a]**
Nivolumab–ipilumumab	First-line treatment of unresectable malignant pleural mesothelioma	A trial of nivolumab in combination with ipilimumab versus pemetrexed with cisplatin or carboplatin as first-line therapy in unresectable pleural mesothelioma (CheckMate743)[2] Phase III NCT02899299	ChT with cisplatin or carboplatin plus pemetrexed Median OS: 14.1 months	OS gain: 4.0 months	OS HR: 0.74 (0.60-0.91)[b]		3 (Form 2a)

Pemetrexed –cisplatin	ChT-naïve unresectable malignant pleural mesothelioma	Study of pemetrexed in combination with cisplatin versus cisplatin alone in patients with malignant pleural mesothelioma (EMPHACIS)[3] Phase III	Cisplatin Median OS: 9.3 months	OS gain: 2.8 months	0.77 PE HR[c]		3 (Form 2a)

CI, confidence interval; ChT, chemotherapy; ESMO-MCBS, European Society for Medical Oncology-Magnitude of Clinical Benefit Scale; HR, hazard ratio; OS, overall survival; PE, point estimate; QoL, quality of life.

[a] ESMO-MCBS v1.1.[4] The scores have been calculated by the ESMO-MCBS Working Group and validated by the ESMO Guidelines Committee (https://www.esmo.org/guidelines/esmo-mcbs/scale-evaluation-forms-v1.0-v1.1/scale-evaluation-forms-v1.1).

[b] HR 96.6% CI reported.

[c] Imputed score assuming the lower limit of the 95% CI was <0.65.

Supplementary Table S4. Levels of evidence and grades of recommendation (adapted from the Infectious Diseases Society of America-United States Public Health Service Grading System[a])

Levels of evidence

I	Evidence from at least one large randomised, controlled trial of good methodological quality (low potential for bias) or meta-analyses of well-conducted randomised trials without heterogeneity
II	Evidence from at least one large randomised, controlled trial of good methodological quality (low potential for bias) or meta-analyses of well-conducted randomised trials without heterogeneity
III	Prospective cohort studies
IV	Retrospective cohort studies or case-control studies
V	Studies without control group, case reports, expert opinions

Grades of recommendation

A	Strong evidence for efficacy with a substantial clinical benefit, strongly recommended
B	Strong or moderate evidence for efficacy but with a limited clinical benefit, generally recommended
C	Insufficient evidence for efficacy or benefit does not outweigh the risk or the disadvantages (adverse events, costs, etc.), optional
D	Moderate evidence against efficacy or for adverse outcome, generally not recommended

E	Strong evidence against efficacy or for adverse outcome, never recommended

[a] Reprinted by permission of Oxford University Press on behalf of the Infectious Diseases Society of America.[5]

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1. TNM classification of malignant tumours, Eighth edition. Chichester, West Sussex, UK ; Hoboken, NJ: John Wiley & Sons, Inc, 2017.

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3. Vogelzang NJ, Rusthoven JJ, Symanowski J et al. Phase III Study of Pemetrexed in Combination With Cisplatin Versus Cisplatin Alone in Patients With Malignant Pleural Mesothelioma. *J Clin Oncol.* 2003; 21(14):2636-2644.

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